Chapter 6

REQUIREMENTS ENGINEERING PROCESS

Requirements Engineering Processes

Processes used to discover, analyse and validate system requirements

Objectives

To describe the principal requirements engineering activities

To introduce techniques for requirements elicitation and analysis

To describe requirements validation

To discuss the role of requirements management in support of other requirements engineering processes

Topics covered

Feasibility studies

Requirements elicitation and analysis

Requirements validation

Requirements management

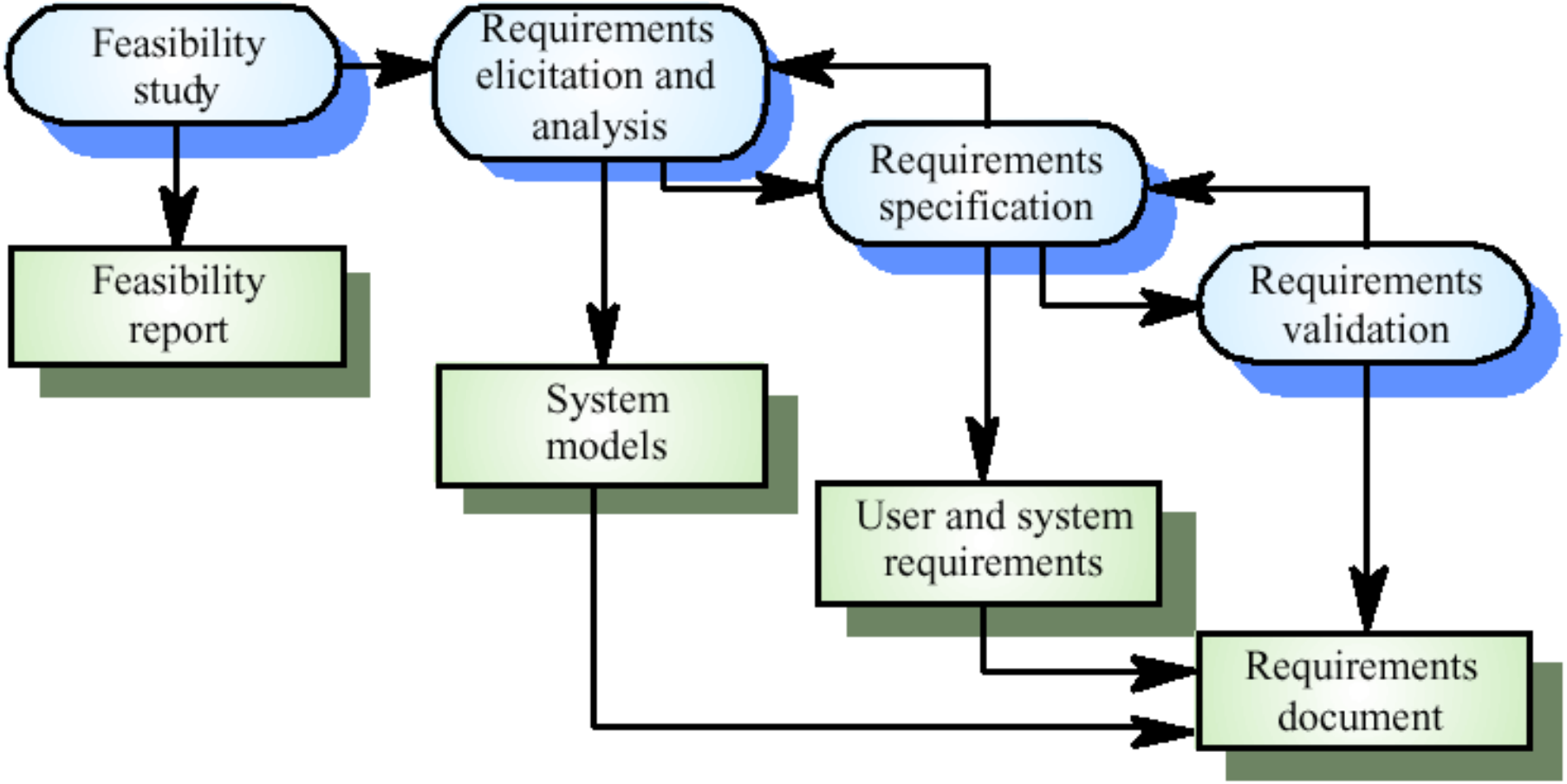
Requirements engineering processes

The processes used for RE vary widely depending on the application domain, the people involved and the organisation developing the requirements

However, there are a number of generic activities common to all processes

- Requirements elicitation
- Requirements analysis
- Requirements validation
- Requirements management

The requirements engineering process



Feasibility studies

A feasibility study decides whether or not the proposed system is worthwhile

A short focused study that checks

- If the system contributes to organisational objectives
- If the system can be engineered using current technology and within budget
- If the system can be integrated with other systems that are used

Feasibility study implementation

Based on information assessment (what is required), information collection and report writing

Questions for people in the organisation

- What if the system wasn't implemented?
- What are current process problems?
- How will the proposed system help?
- What will be the integration problems?
- Is new technology needed? What skills?
- What facilities must be supported by the proposed system?

Elicitation and analysis

Sometimes called requirements elicitation or requirements discovery

Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints

May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called *stakeholders*

Problems of requirements analysis

Stakeholders don’t know what they really want

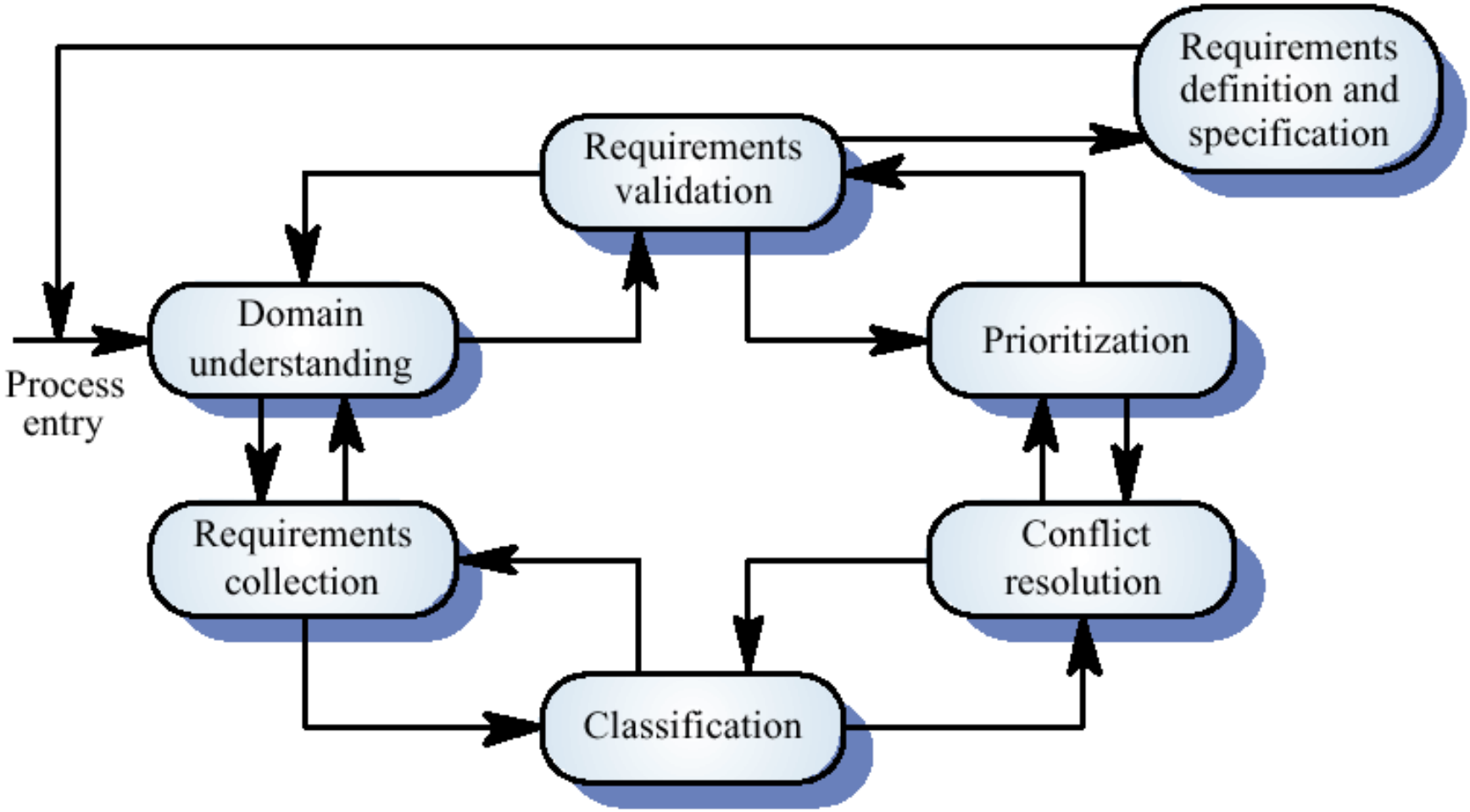
Stakeholders express requirements in their own terms

Different stakeholders may have conflicting requirements

Organisational and political factors may influence the system requirements

The requirements change during the analysis process. New stakeholders may emerge and the business environment change

The requirements analysis process



Process activities

Domain understanding

Requirements collection

Classification

Conflict resolution

Prioritisation

Requirements checking

System models

Different models may be produced during the requirements analysis activity

Requirements analysis may involve three structuring activities which result in these different models

- Partitioning. Identifies the structural (part-of) relationships between entities
- Abstraction. Identifies generalities among entities
- Projection. Identifies different ways of looking at a problem

System models covered in Chapter 7

Viewpoint-oriented elicitation

Stakeholders represent different ways of looking at a problem or problem viewpoints

This multi-perspective analysis is important as there is no single correct way to analyse system requirements

Banking ATM system

The example used here is an auto-teller system which provides some automated banking services

I use a very simplified system which offers some services to customers of the bank who own the system and a narrower range of services to other customers

Services include cash withdrawal, message passing (send a message to request a service), ordering a statement and transferring funds

Autoteller viewpoints

Bank customers

Representatives of other banks

Hardware and software maintenance engineers

Marketing department

Bank managers and counter staff

Database administrators and security staff

Communications engineers

Personnel department

Types of viewpoint

Data sources or sinks

- Viewpoints are responsible for producing or consuming data. Analysis involves checking that data is produced and consumed and that assumptions about the source and sink of data are valid

Representation frameworks

- Viewpoints represent particular types of system model. These may be compared to discover requirements that would be missed using a single representation. Particularly suitable for real-time systems

Receivers of services

- Viewpoints are external to the system and receive services from it. Most suited to interactive systems

External viewpoints

Natural to think of end-users as receivers of system services

Viewpoints are a natural way to structure requirements elicitation

It is relatively easy to decide if a viewpoint is valid

Viewpoints and services may be sued to structure non-functional requirements

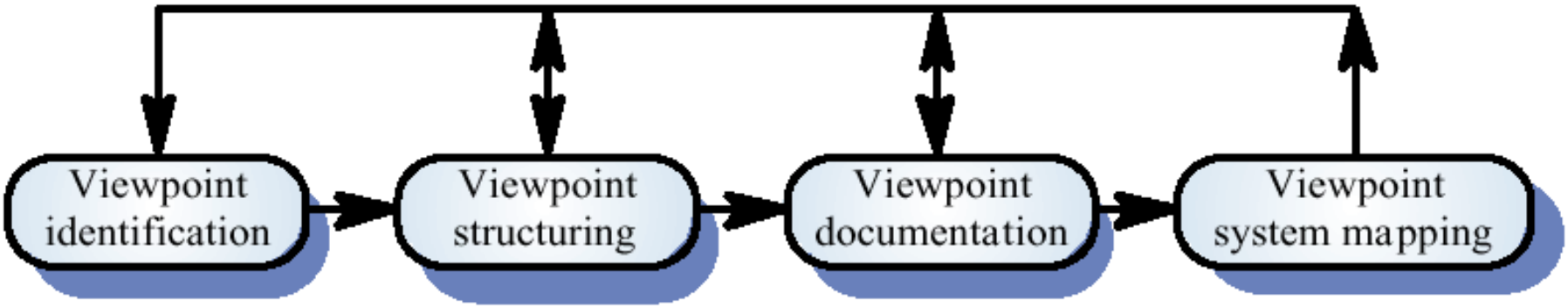
Method-based analysis

Widely used approach to requirements analysis. Depends on the application of a structured method to understand the system

Methods have different emphases. Some are designed for requirements elicitation, others are close to design methods

A viewpoint-oriented method (VORD) is used as an example here. It also illustrates the use of viewpoints

The VORD method



VORD process model

Viewpoint identification

- Discover viewpoints which receive system services and identify the services provided to each viewpoint

Viewpoint structuring

- Group related viewpoints into a hierarchy. Common services are provided at higher-levels in the hierarchy

Viewpoint documentation

- Refine the description of the identified viewpoints and services

Viewpoint-system mapping

- Transform the analysis to an object-oriented design

VORD standard forms

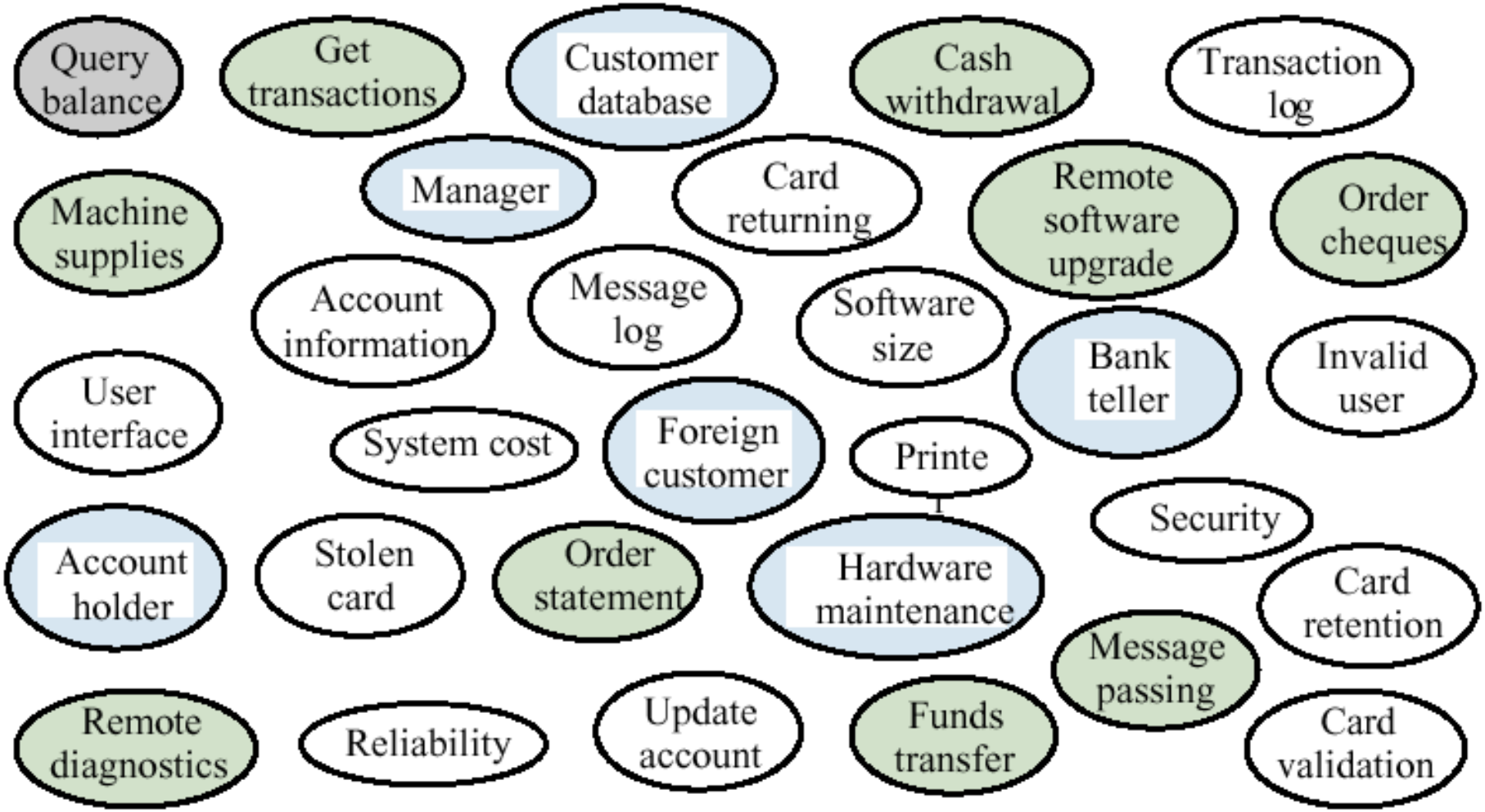
Viewpoint template

Reference:	The viewpoint name.
Attributes:	Attributes providing viewpoint information.
Events:	A reference to a set of event scenarios describing how the system reacts to viewpoint events.
Services	A reference to a set of service descriptions.
Sub-VPs:	The names of sub-viewpoints.

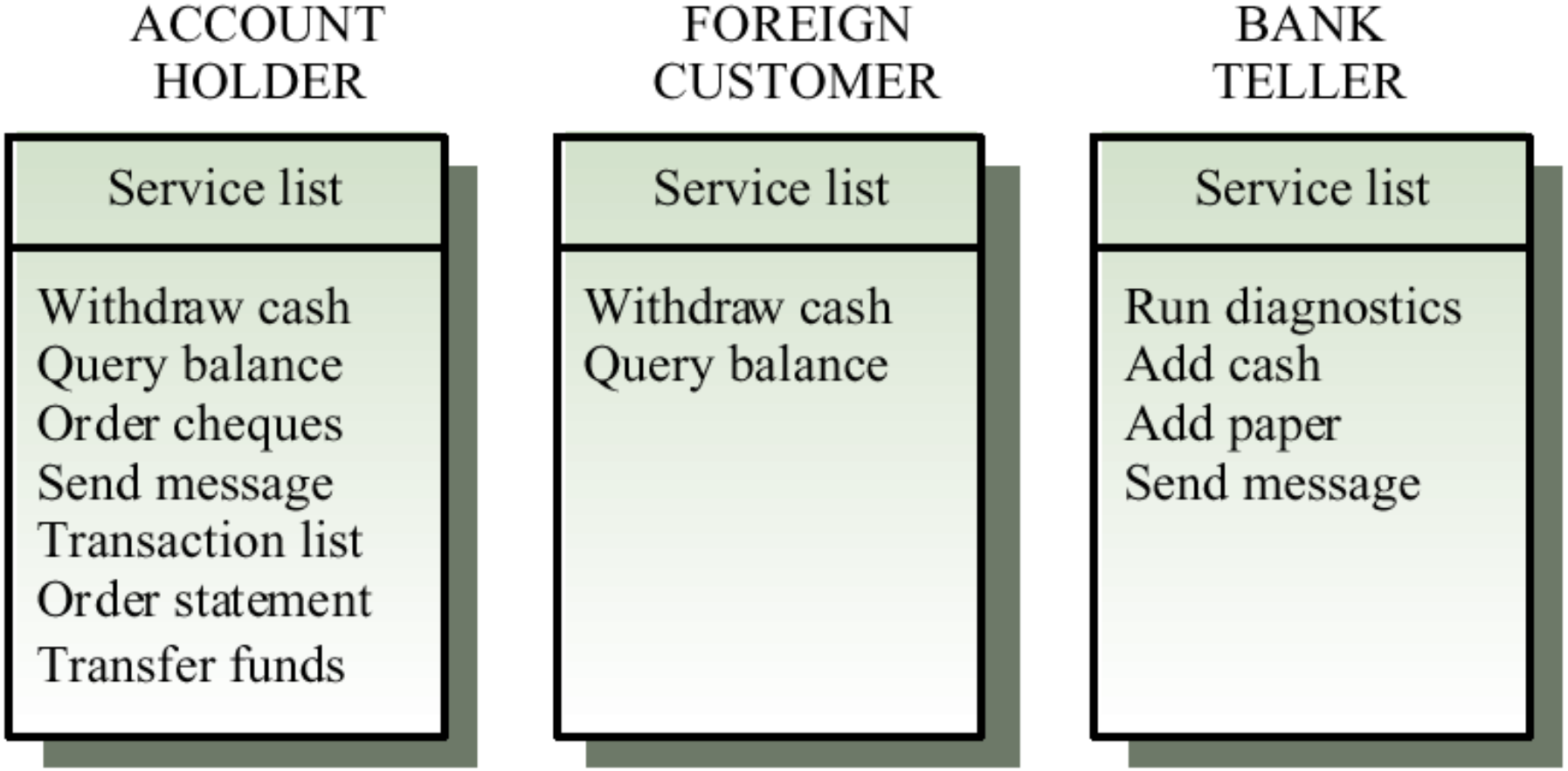
Service template

Reference:	The service name.
Rationale:	Reason why the service is provided.
Specification:	Reference to a list of service specifications. These may be expressed in different notations.
Viewpoints:	List of viewpoint names receiving the service.
Non-functional requirements:	Reference to a set of non-functional requirements which constrain the service.
Provider:	Reference to a list of system objects which provide the service.

Viewpoint identification



Viewpoint service information

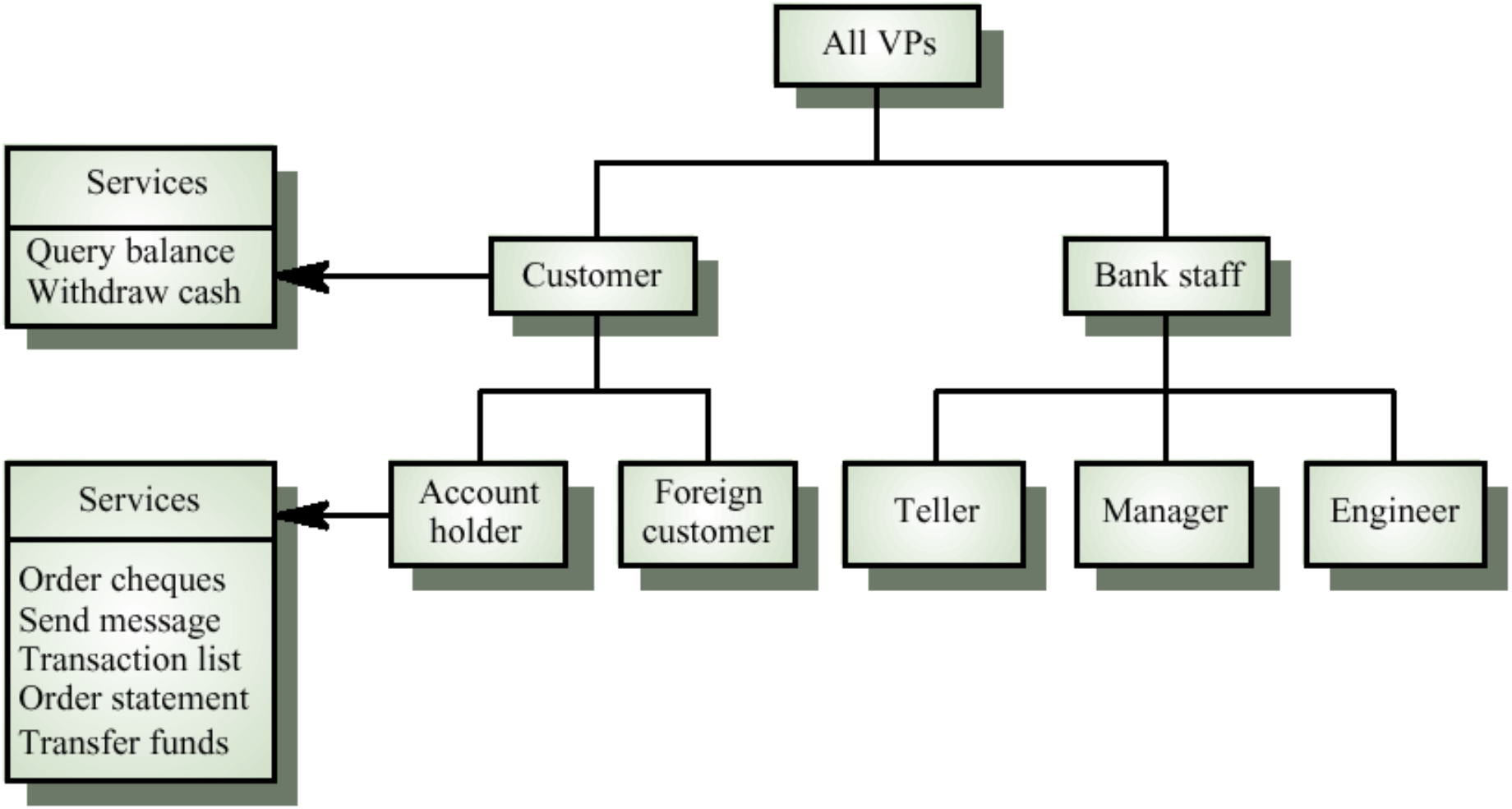


Viewpoint data/control

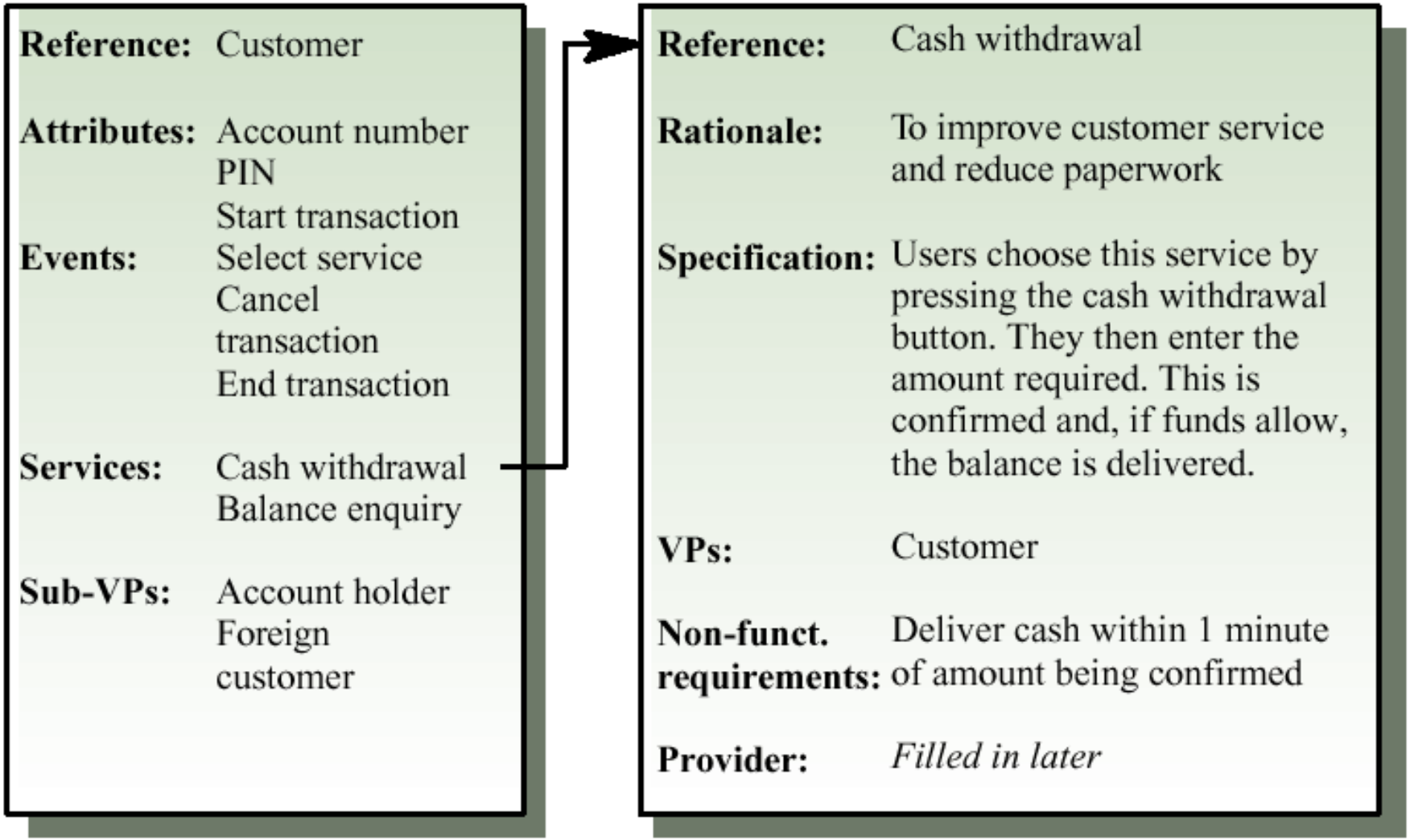
ACCOUNT HOLDER

Control input	Data input
Start transaction Cancel transaction End transaction Select service	Card details PIN Amount required Message

Viewpoint hierarchy



Customer/cash withdrawal templates



Scenarios

Scenarios are descriptions of how a system is used in practice

They are helpful in requirements elicitation as people can relate to these more readily than abstract statement of what they require from a system

Scenarios are particularly useful for adding detail to an outline requirements description

Scenario descriptions

System state at the beginning of the scenario

Normal flow of events in the scenario

What can go wrong and how this is handled

Other concurrent activities

System state on completion of the scenario

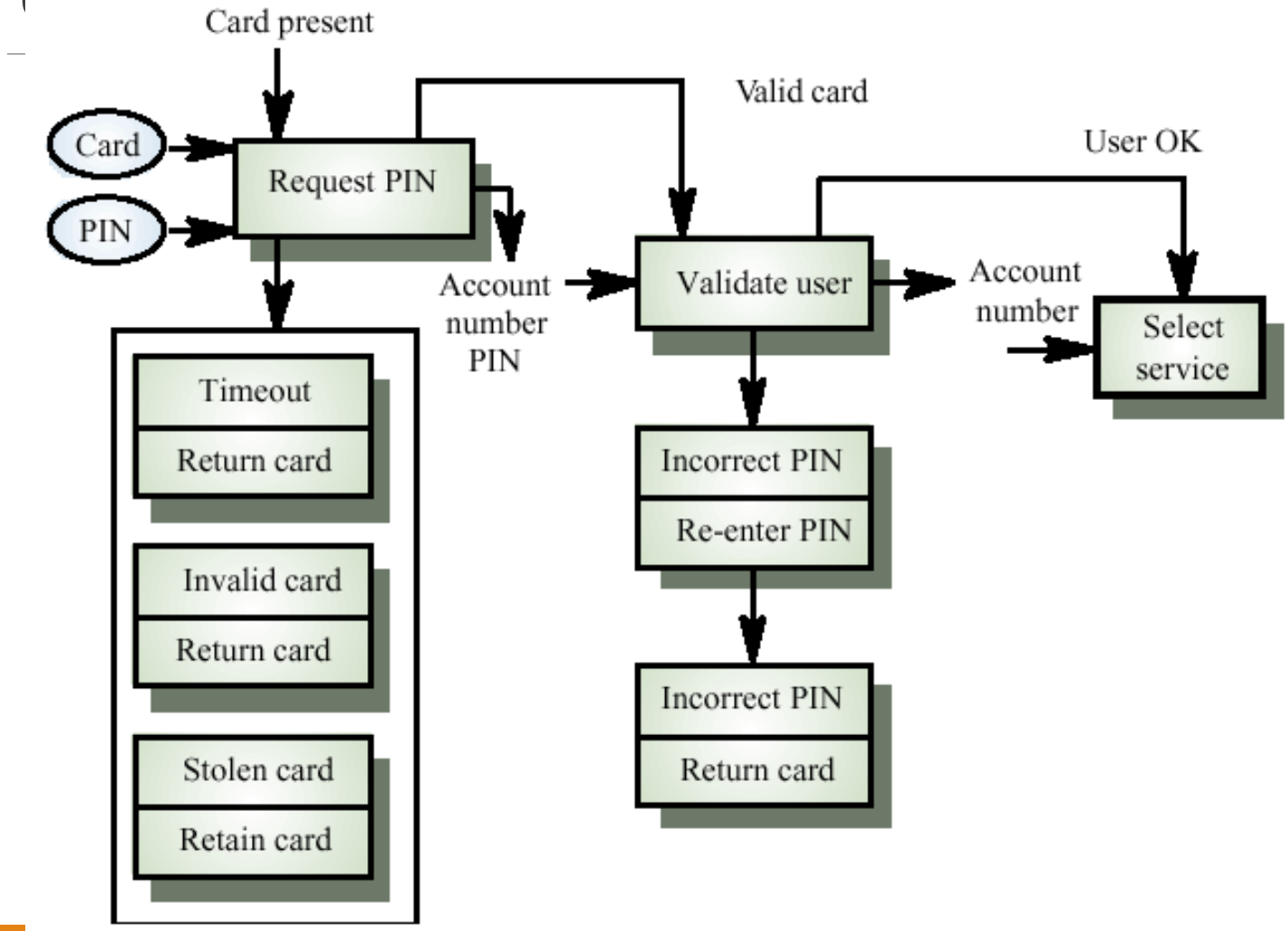
Event scenarios

Event scenarios may be used to describe how a system responds to the occurrence of some particular event such as ‘start transaction’

VORD includes a diagrammatic convention for event scenarios.

- Data provided and delivered
- Control information
- Exception processing
- The next expected event

Event scenario - start transaction



Notation for data and control analysis

Ellipses. data provided from or delivered to a viewpoint

Control information enters and leaves at the top of each box

Data leaves from the right of each box

Exceptions are shown at the bottom of each box

Name of next event is in box with thick edges

Exception description

Most methods do not include facilities for describing exceptions

In this example, exceptions are

- Timeout. Customer fails to enter a PIN within the allowed time limit
- Invalid card. The card is not recognised and is returned
- Stolen card. The card has been registered as stolen and is retained by the machine

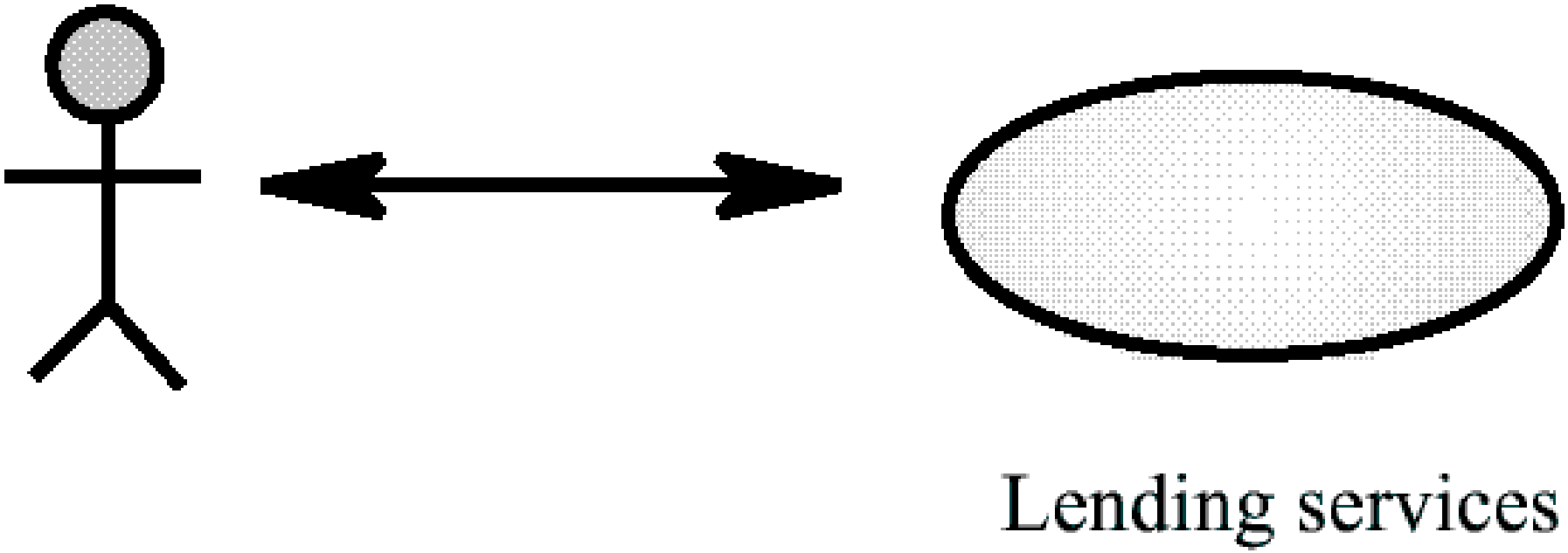
Use cases

Use-cases are a scenario based technique in the UML which identify the actors in an interaction and which describe the interaction itself

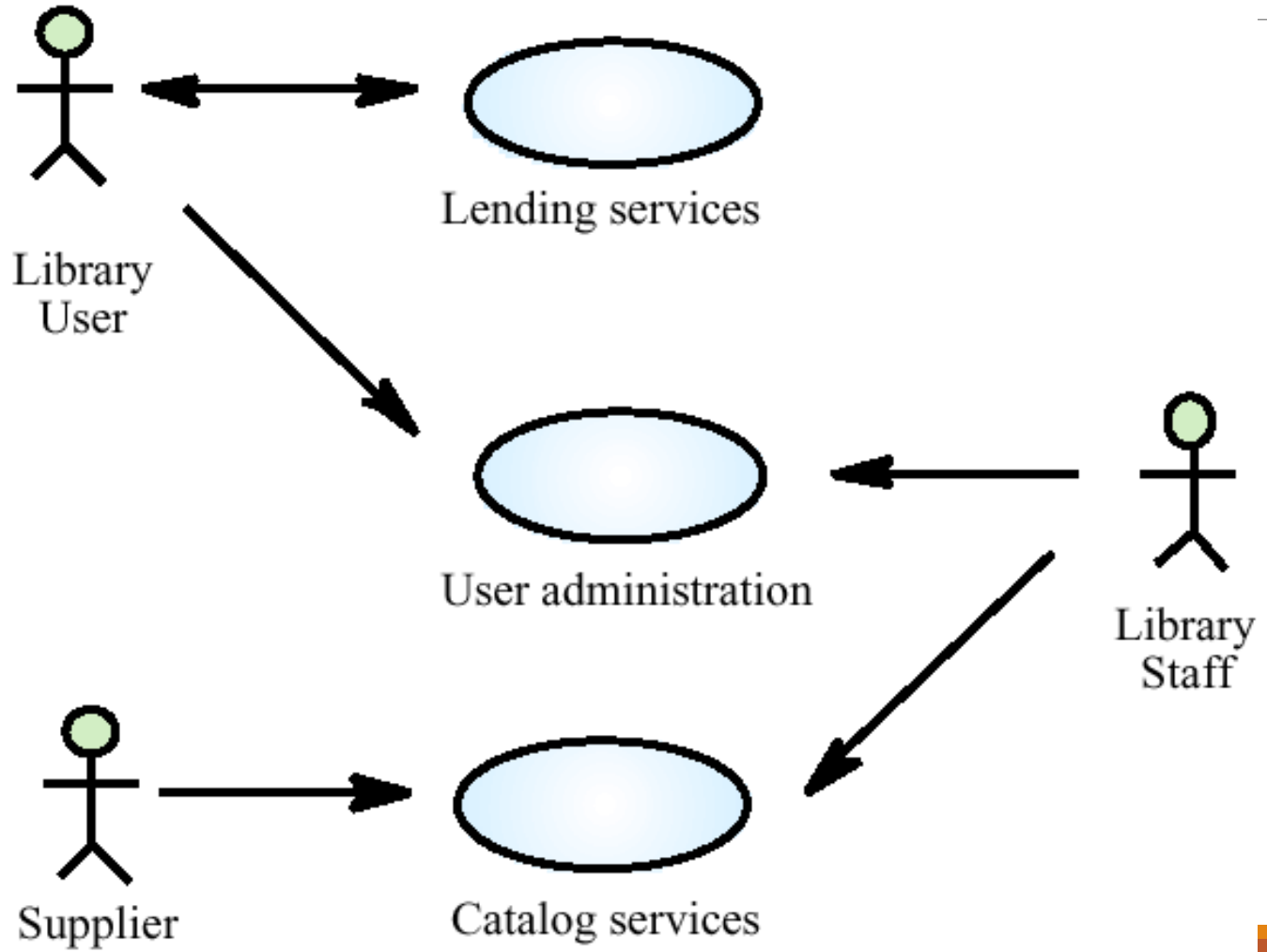
A set of use cases should describe all possible interactions with the system

Sequence diagrams may be used to add detail to use-cases by showing the sequence of event processing in the system

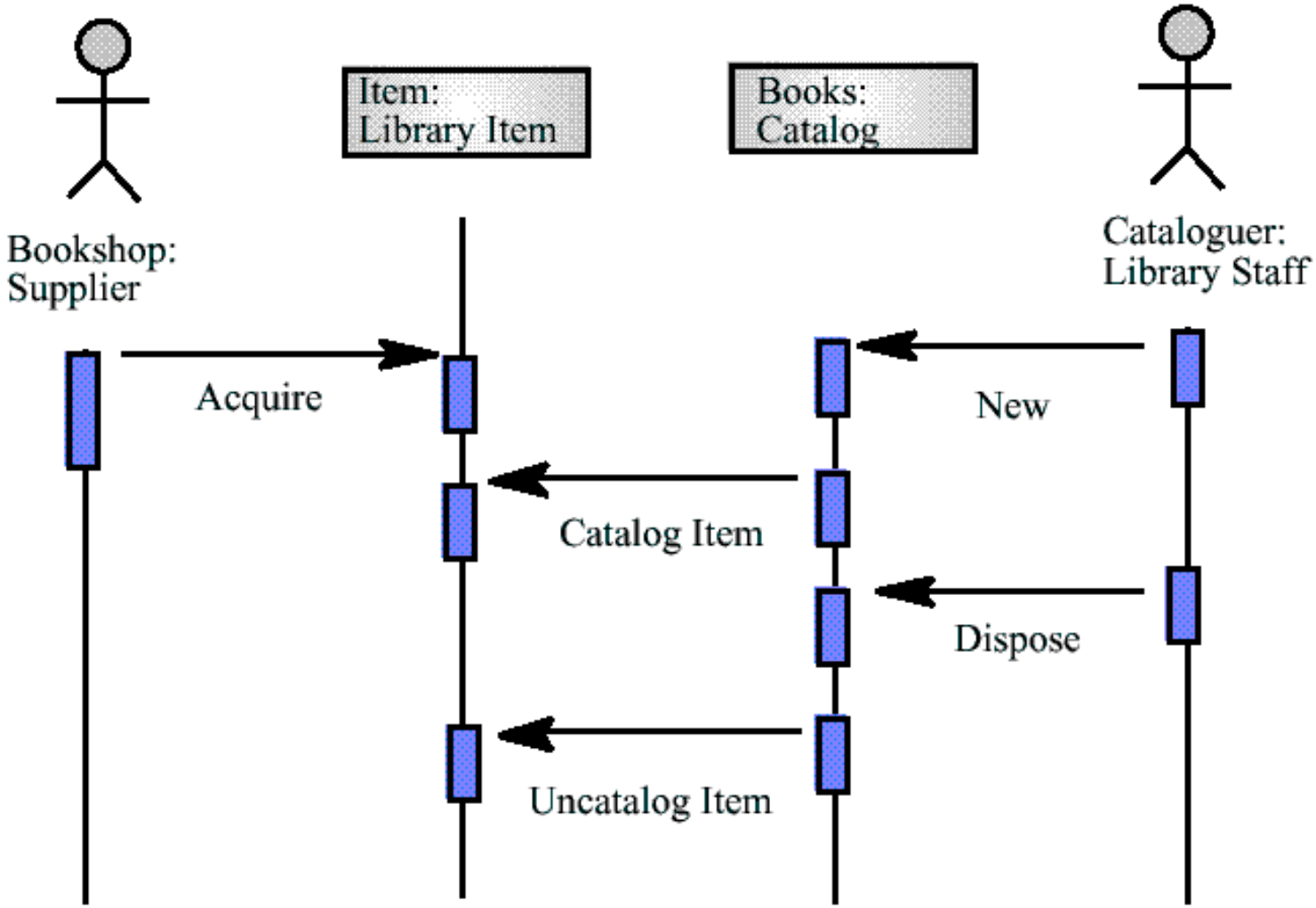
Lending use-case



Library use-cases



Catalogue management



Social and organisational factors

Software systems are used in a social and organisational context. This can influence or even dominate the system requirements

Social and organisational factors are not a single viewpoint but are influences on all viewpoints

Good analysts must be sensitive to these factors but currently no systematic way to tackle their analysis

Example

Consider a system which allows senior management to access information without going through middle managers

- Managerial status. Senior managers may feel that they are too important to use a keyboard. This may limit the type of system interface used
- Managerial responsibilities. Managers may have no uninterrupted time where they can learn to use the system
- Organisational resistance. Middle managers who will be made redundant may deliberately provide misleading or incomplete information so that the system will fail

Ethnography

A social scientists spends a considerable time observing and analysing how people actually work

People do not have to explain or articulate their work

Social and organisational factors of importance may be observed

Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models

Focused ethnography

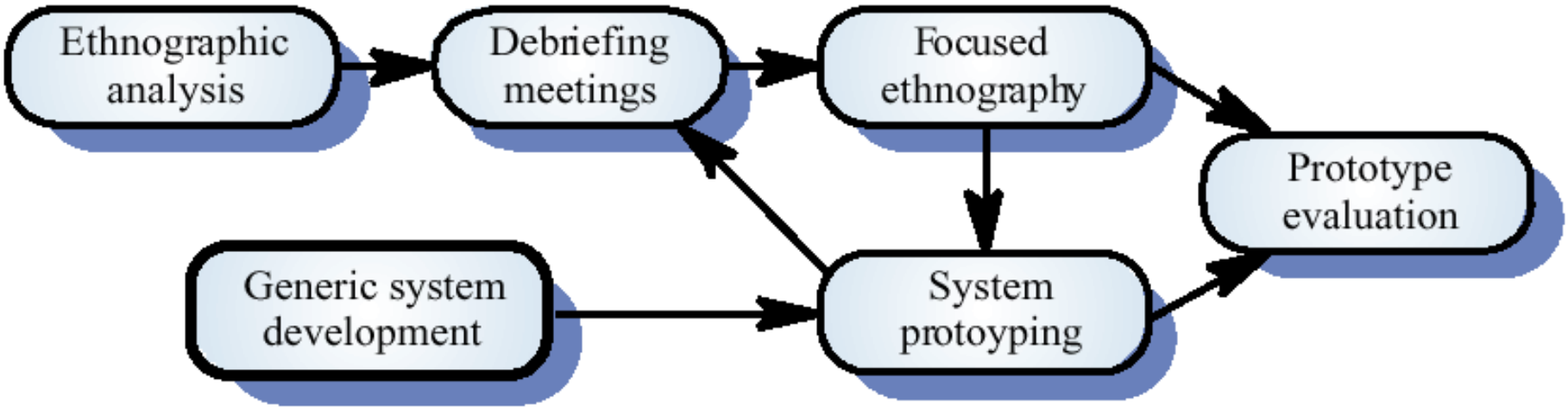
Developed in a project studying the air traffic control process

Combines ethnography with prototyping

Prototype development results in unanswered questions which focus the ethnographic analysis

Problem with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant

Ethnography and prototyping



Scope of ethnography

Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work

Requirements that are derived from cooperation and awareness of other people's activities

Requirements validation

Concerned with demonstrating that the requirements define the system that the customer really wants

Requirements error costs are high so validation is very important

- Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error

Requirements checking

Validity. Does the system provide the functions which best support the customer's needs?

Consistency. Are there any requirements conflicts?

Completeness. Are all functions required by the customer included?

Realism. Can the requirements be implemented given available budget and technology

Verifiability. Can the requirements be checked?

Requirements validation techniques

Requirements reviews
- Systematic manual analysis of the requirements

Prototyping
- Using an executable model of the system to check requirements. Covered in Chapter 8

Test-case generation
- Developing tests for requirements to check testability

Automated consistency analysis
- Checking the consistency of a structured requirements description

Requirements reviews

Regular reviews should be held while the requirements definition is being formulated

Both client and contractor staff should be involved in reviews

Reviews may be formal (with completed documents) or informal. Good communications between developers, customers and users can resolve problems at an early stage

Review checks

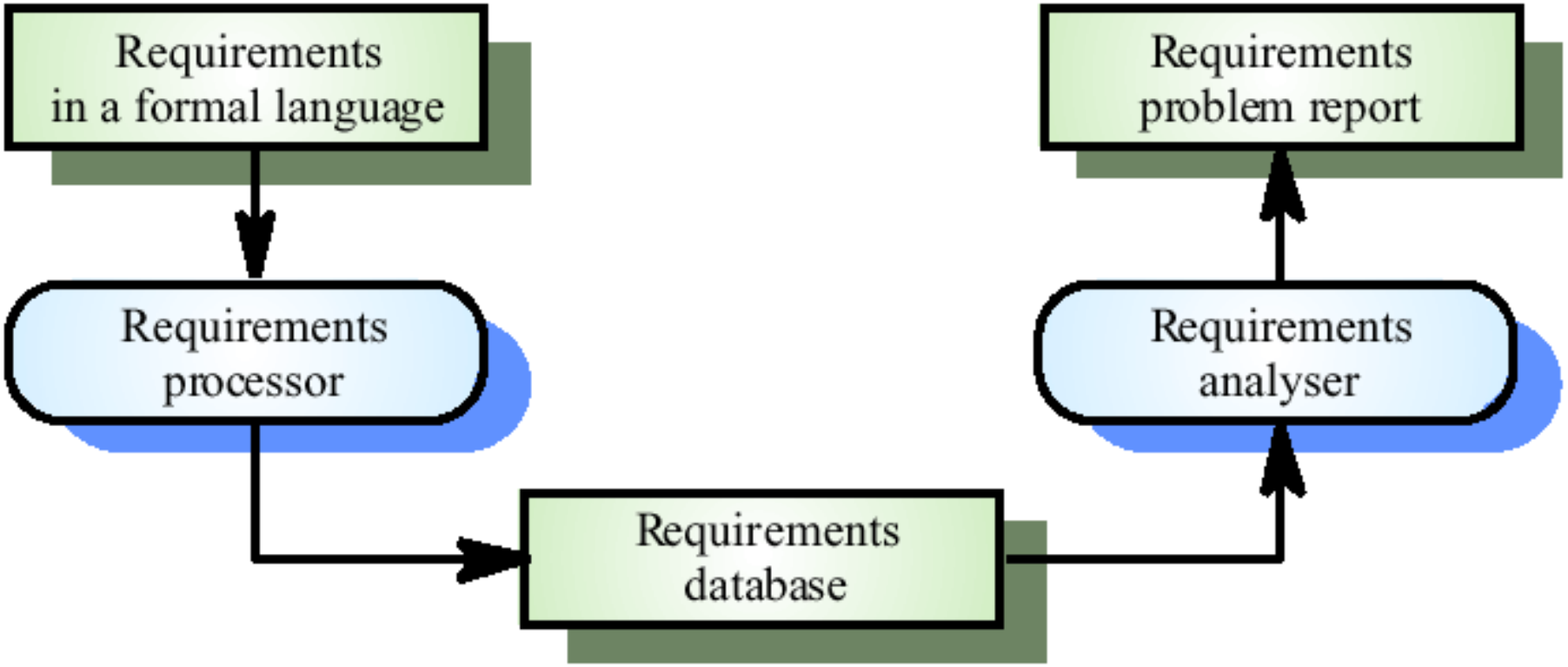
Verifiability. Is the requirement realistically testable?

Comprehensibility. Is the requirement properly understood?

Traceability. Is the origin of the requirement clearly stated?

Adaptability. Can the requirement be changed without a large impact on other requirements?

Automated consistency checking



Requirements management

Requirements management is the process of managing changing requirements during the requirements engineering process and system development

Requirements are inevitably incomplete and inconsistent

- New requirements emerge during the process as business needs change and a better understanding of the system is developed
- Different viewpoints have different requirements and these are often contradictory

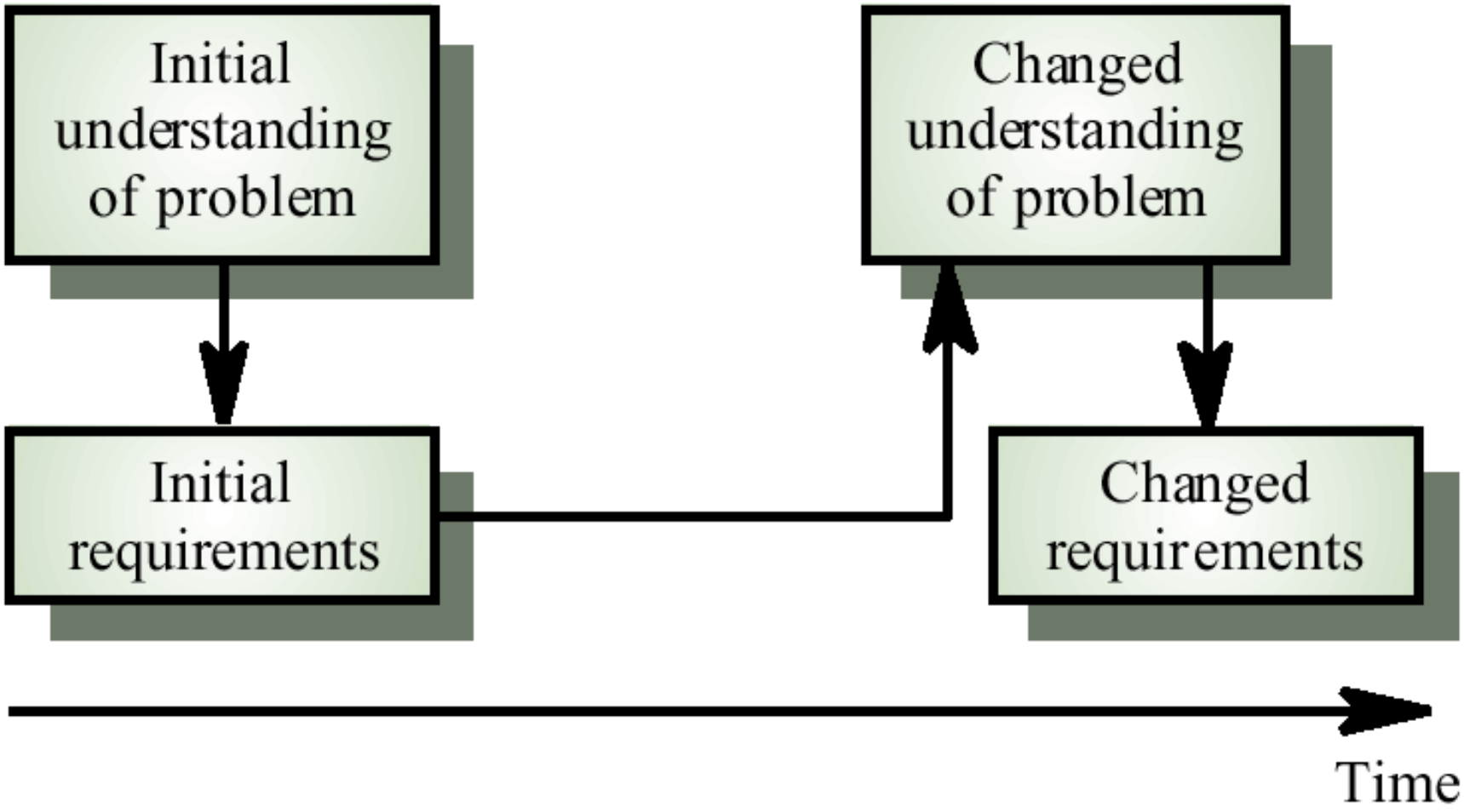
Requirements change

The priority of requirements from different viewpoints changes during the development process

System customers may specify requirements from a business perspective that conflict with end-user requirements

The business and technical environment of the system changes during its development

Requirements evolution



Enduring and volatile requirements

Enduring requirements. Stable requirements derived from the core activity of the customer organisation. E.g. a hospital will always have doctors, nurses, etc. May be derived from domain models

Volatile requirements. Requirements which change during development or when the system is in use. In a hospital, requirements derived from health-care policy

Classification of requirements

Mutable requirements

- Requirements that change due to the system's environment

Emergent requirements

- Requirements that emerge as understanding of the system develops

Consequential requirements

- Requirements that result from the introduction of the computer system

Compatibility requirements

- Requirements that depend on other systems or organisational processes

Requirements management planning

During the requirements engineering process, you have to plan:

- Requirements identification
  - How requirements are individually identified
- A change management process
  - The process followed when analysing a requirements change
- Traceability policies
  - The amount of information about requirements relationships that is maintained
- CASE tool support
  - The tool support required to help manage requirements change

Traceability

Traceability is concerned with the relationships between requirements, their sources and the system design

Source traceability

- Links from requirements to stakeholders who proposed these requirements

Requirements traceability

- Links between dependent requirements

Design traceability

- Links from the requirements to the design

A traceability matrix

Req. id	1.1	1.2	1.3	2.1	2.2	2.3	3.1	3.2
1.1		U	R					
1.2			U			R		U
1.3	R			R				
2.1			R		U			U
2.2								U
2.3		R		U				
3.1								R
3.2							R	

CASE tool support

Requirements storage

- Requirements should be managed in a secure, managed data store

Change management

- The process of change management is a workflow process whose stages can be defined and information flow between these stages partially automated

Traceability management

- Automated retrieval of the links between requirements

Requirements change management

Should apply to all proposed changes to the requirements

Principal stages

- Problem analysis. Discuss requirements problem and propose change
- Change analysis and costing. Assess effects of change on other requirements
- Change implementation. Modify requirements document and other documents to reflect change

Requirements change management



Key points

The requirements engineering process includes a feasibility study, requirements elicitation and analysis, requirements specification and requirements management

Requirements analysis is iterative involving domain understanding, requirements collection, classification, structuring, prioritisation and validation

Systems have multiple stakeholders with different requirements

Key points

Social and organisation factors influence system requirements

Requirements validation is concerned with checks for validity, consistency, completeness, realism and verifiability

Business changes inevitably lead to changing requirements

Requirements management includes planning and change management